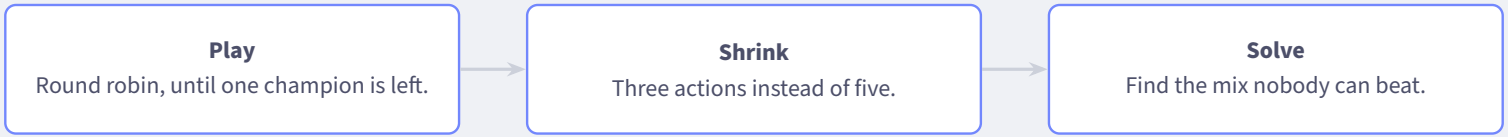
Nash equilibrium

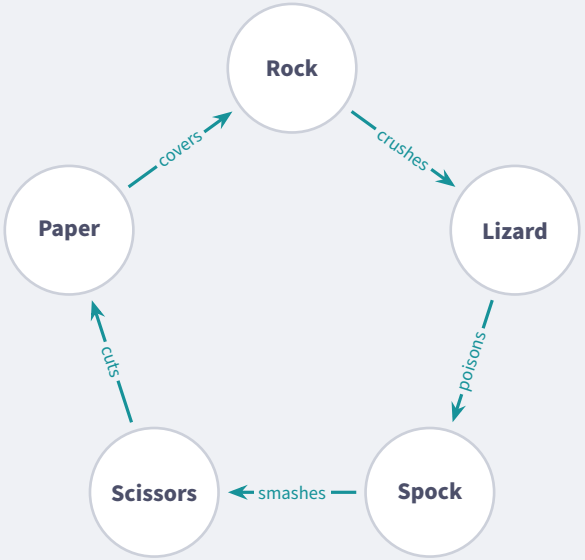
Rock Paper Scissors Lizard Spock, all the way down



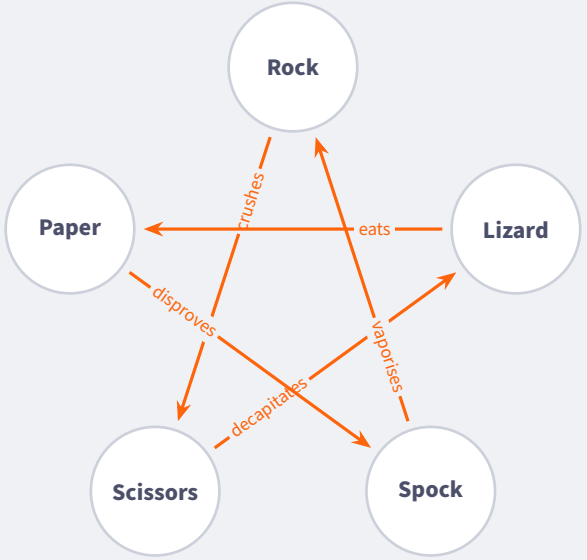
Class champion:

Did I beat them?

Beat the one beside you



Beat the one across from you



Each action wins ____, loses ____ and draws ____. So the game is _____, and every cell's payoffs add to ____.

	Rock	Paper	Scissors
Rock			
Paper			
Scissors			

A cell with both players' best responses circled would be a pure Nash equilibrium.

Fill in the *row* player's payoff:

win = +1, draw = 0, lose = −1.

Then circle each player's best responses.

The column player's matrix is .

Whatever the pair of actions, somebody wants to move:

If they play Rock, I want

If they play Paper, I want

If they play Scissors, I want

… and then they want to move too.

So any equilibrium has to be _____.

A mixed strategy σ is a best response to τ if and only if

In words: every action I actually use has to be _____.

So if I mix over all three actions, all three must give me the same expected payoff.

Suppose the column player uses $\sigma_c = (x, y, 1 - x - y)$ over (Rock, Paper, Scissors).

$u_r(\text{Rock}, \sigma_c) =$

$u_r(\text{Paper}, \sigma_c) =$

$u_r(\text{Scissors}, \sigma_c) =$

Set them equal, and solve:

$\sigma_c =$

Support enumeration checks every pair of supports in turn.

Non-empty supports for one player, out of 3 actions:

Pairs of supports to check:

And with 5 actions, as in the game we played:

Which is why we want theorems, not enumeration.

What we just did in the room is **Question 1** on the Nash Equilibrium page.

(a)	Why it is symmetric and zero sum, and the column player's matrix.	[3]
(b)	Utilities for three given pairs of mixed strategies.	[3]
(c)	Use the best response condition to prove $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is an equilibrium.	[5]
(d)	Prove it is the *unique* equilibrium.	[5]
(e)	State and justify the equilibrium for the five-action game.	[3]

vknight.org/gt/topics/nash-equilibrium.html

Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
Five actions, then three	2.2.3	Games: Definition: Normal Form Game
Nobody wants to stand still	5.2.1	Nash Equilibrium: Definition: Nash Equilibria
Equal payoffs across the support	3.2.6	Rationalisation: Theorem: Best Response Condition
Checking every pair of supports	5.2.3	Nash Equilibrium: Definition: Support Enumeration Algorithm
Examinable	3.2.6	Theorem: Best Response Condition, and proof

vknight.org/gtb/chapters/rationalisation, vknight.org/gtb/chapters/nash-equilibrium

Quick quiz before you go: vknight.org/gt/topics/nash-equilibrium.html

Before the next class: read the Nash Equilibrium chapter as far as support enumeration, and attempt Question 1.